

Medical Alert Interrupt Latency Demonstration

EEE 4775 Real Time Systems Capstone

Project summary: A simulated medical patient call system that compares binary semaphore and direct task notification wake latency under idle, loaded, and fault injected conditions.

System Architecture

- GPIO 18 button represents a patient call event and triggers a falling edge interrupt.
- The ISR records a timestamp, pulses GPIO 19 for logic analyzer timing, and wakes two bottom half tasks.
- One bottom half uses a binary semaphore while the other uses a direct task notification.
- Four periodic workloads create controlled contention on Core 1 using rate monotonic priorities.
- Task A (priority 15) can preempt the priority 12 bottom halves while task B-D cannot.

Operating Modes

- `'WITH_LOAD = 0'`: baseline run without the four additional workloads causing contention.
- `'WITH_LOAD = 1'`: the four periodic background tasks are enabled.
- `'INJECT_MISSING_ISR_YIELD = 1'`: omits `'portYIELD_FROM_ISR()'` to demonstrate delayed task scheduling.

Key Concepts

- ISR to task deferral and ISR safe FreeRTOS APIs
- Binary semaphore versus direct task notification trade offs
- Rate monotonic priority assignment, WCET observation, task heartbeats, and fault injection.
- Logic analyzer/VCD evidence for interrupt response timing.

Measured Results (100 button presses)

Configuration	Notif.	Sem.
No load, normal yield	30 μ s	2604 μ s
Loaded, normal yield	2450 μ s	2646 μ s
No load, yield removed	2279 μ s	2629 μ s
Loaded, yield removed	2467 μ s	2728 μ s

Interpretation of Results

- The direct task notification produced the lowest measured worst-case latency in both normal operating modes.
- The notification path increases from 30 μ s to 2450 μ s under load because higher priority Task A could delay the bottom half.
- Removing the ISR exit yield increased latency, but the effect was smaller than expected in the loaded run.
- Representative VCD captures measured 15.35 μ s from the GPIO falling edge to GPIO 19 rising edge in both idle and loaded cases.

Known Limitations

- The periodic workloads were reduced in weight to prevent repeated watchdog resets.
- Reported load task timing may include preemption and is best described as maximum observed duration rather than an isolated CPU execution time.
- The project is a simulation and timing demonstration, not a production medical device.
- Possible risk of multiple button presses merging into a single event for the binary semaphore whereas task notifications are incremented.

Portfolio Contents

- Wokwi project link and demo video.
- Architecture diagram, task/WCET table, and hazard analysis within the README, supporting work pdf, and final reflection.